

A MEMBRANE-DISRUPTIVE ACTION OF VBIT-4 CHALLENGES ITS ROLE AS A WIDELY USED VDAC OLIGOMERIZATION INHIBITOR

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ABSTRACT

The voltage-dependent anion channel (VDAC) is the most abundant protein of the mitochondrial outer membrane and plays a central role in metabolism and apoptosis. Its oligomerization has been proposed to mediate critical processes, including mtDNA release and apoptotic signaling. The small molecule VBIT-4 has been widely used over the past decade as an inhibitor of VDAC1 oligomerization, yet its molecular mechanisms remain poorly defined. Here, we combined high-speed atomic force microscopy, single-channel electrophysiology, fluorescence spectroscopy, microscale thermophoresis, and molecular dynamics simulations to investigate its mode of action. We show that VBIT-4 partitions into lipid bilayers with micromolar affinity, where it perturbs lipid packing, decreases bilayer thickness, and induces lipid flip-flop and pore-like defects. AFM directly visualized VBIT-4 localizing to protein-free lipid regions, leaving VDAC1 clusters unaffected, while electrophysiology confirmed membrane permeabilization at concentrations ≥ 20 μ M. These membrane-disruptive effects occurred independently of VDAC1 and were accompanied by VDAC1-independent cytotoxicity in HeLa cells, including mitochondrial depolarization and impaired calcium homeostasis. Importantly, VBIT-4's ability to alter lipid organization, provides an explanation for previously reported reductions in VDAC cross-linking, which likely arise from lipid-driven rearrangements of VDAC clusters rather than inhibition of discrete protein-protein oligomers. Our findings establish that VBIT-4 is not a specific inhibitor of VDAC oligomerization. Rather, it is a membrane-active compound whose effects result from the modulation of membrane-dependent protein interactions. This work further reveals the risk of broad off-target effects, calling for caution in interpreting prior studies that have employed VBIT-4 and underscores the necessity of using lipid-based controls when evaluating small molecules that target membrane proteins.